

11장. 미션 & 연습문제

미션

In [1]:

```
#미션
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

df = pd.read_excel('사과배복숭아-학습.xlsx')
df.fillna(0, inplace=True)
df.replace({'fruit':{'사과':0, '배':1, '복숭아':2}}, inplace=True)
x_train = df.iloc[:, 1:]
y_train = df.iloc[:,0]

model_3 = RandomForestClassifier(n_estimators=3,random_state=0)
model_3.fit(x_train, y_train)

model_9 = RandomForestClassifier(n_estimators=9,random_state=0)
model_9.fit(x_train, y_train)

model_15 = RandomForestClassifier(n_estimators=15,random_state=0)
model_15.fit(x_train, y_train)

df_test = pd.read_excel('사과배복숭아-테스트.xlsx')
df_test.fillna(0, inplace=True)
df_test.replace({'fruit':{'사과':0, '배':1, '복숭아':2}}, inplace=True)
x_test = df_test.iloc[:, 1:]
y_test = df_test.iloc[:,0]

prd_3 = model_3.predict(x_test)
prd_9 = model_9.predict(x_test)
prd_15 = model_15.predict(x_test)

# 예측값과 테스트 데이터의 정답을 비교해서 정확도 계산
print('정확도(트리 3개) =',accuracy_score(y_test, prd_3))
print('정확도(트리 9개) =',accuracy_score(y_test, prd_9))
print('정확도(트리 15개) =',accuracy_score(y_test, prd_15))
```

정확도(트리 3개) = 0.9
정확도(트리 9개) = 0.9
정확도(트리 15개) = 1.0

연습문제

2번

In [2]:

```
import pandas as pd
import numpy as np
df = pd.read_excel('사과배복숭아-학습.xlsx')
print("결측치 처리 전")
print(df.isna().sum())

# 연습문제2
df.fillna(0, inplace=True)

print("\n결측치 처리 후")
print(df.isna().sum())
```

```
결측치 처리 전
fruit      0
weight     0
height     0
width      0
hardness   0
sweet      0
sour       0
color     49
dtype: int64
```

```
결측치 처리 후
fruit      0
weight     0
height     0
width      0
hardness   0
sweet      0
sour       0
color      0
dtype: int64
```

3번

In [3]:

```
import pandas as pd
import numpy as np
df = pd.read_excel('사과배복숭아-학습.xlsx')
df.fillna(0, inplace=True)
df.replace({'fruit':{'사과':0, '배':1, '복숭아':2}}, inplace=True)
x_train = df.iloc[:, 1:] # 학습 데이터의 속성값
y_train = df.iloc[:,0]   # 학습 데이터의 정답

# 연습문제3
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(random_state=0)
model.fit(x_train, y_train)
```

Out[3]:

DecisionTreeClassifier(random_state=0)

5번

In [4]:

```
import pandas as pd
import numpy as np
df = pd.read_excel('사과배복숭아-학습.xlsx')
df.fillna(0, inplace=True)
df.replace({'fruit':{'사과':0, '배':1, '복숭아':2}}, inplace=True)
x_train = df.iloc[:, 1:] # 학습 데이터의 속성값
y_train = df.iloc[:,0]   # 학습 데이터의 정답

# 연습문제5
from sklearn.neighbors import KNeighborsClassifier

model = KNeighborsClassifier(n_neighbors=3)
model.fit(x_train, y_train)
```

Out[4]:

KNeighborsClassifier(n_neighbors=3)

6번

In [5]:

```
import pandas as pd
import numpy as np
df = pd.read_excel('사과배복숭아-학습.xlsx')
df.fillna(0, inplace=True)
df.replace({'fruit':{'사과':0, '배':1, '복숭아':2}}, inplace=True)
x_train = df.iloc[:, 1:] # 학습 데이터의 속성값
y_train = df.iloc[:,0]   # 학습 데이터의 정답
df_test = pd.read_excel('사과배복숭아-테스트.xlsx')
df_test.fillna(0, inplace=True)
df_test.replace({'fruit':{'사과':0, '배':1, '복숭아':2}}, inplace=True)
x_test = df_test.iloc[:, 1:]
y_test = df_test.iloc[:,0]

from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(random_state=0)
model.fit(x_train, y_train)

# 연습문제6
from sklearn.metrics import accuracy_score
prd = model.predict(x_test)
print('정확도 =', accuracy_score(y_test, prd))
```

정확도 = 0.9